

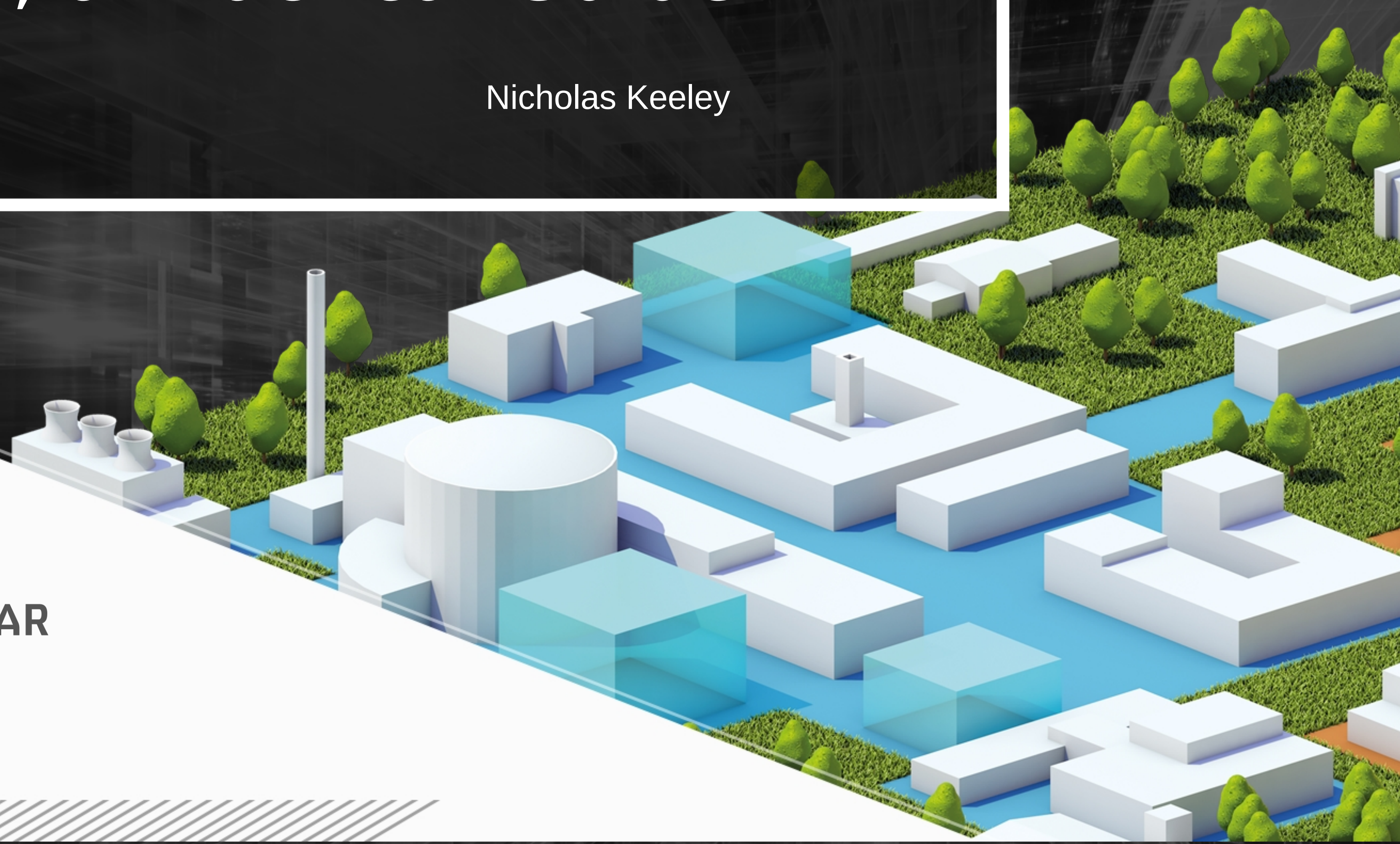


Direct Reactions With FRESCO; a Practical Guide

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In this second part we will look at how to perform transfer reaction calculations with FRESCO. We begin with the distorted wave Born approximation or DWBA

There are two main forms of DWBA theory, zero-range and full finite-range. We will discuss the latter only, since with modern computers there is no need to use the less accurate zero-range formulation

What, then, are the ingredients of a typical DWBA calculation?

DWBA treats the transfer step as a perturbation of the elastic scattering in entrance and exit channels. We therefore need optical potentials at the appropriate energies for entrance and exit channels to calculate the corresponding distorted waves

What do we mean by “appropriate” energies? For the entrance channel, simply the beam energy of the transfer measurement; the distorted waves must describe the elastic scattering of the beam from the target at this energy.

For the exit channel we need an optical potential that describes the elastic scattering of the ejectile from the residual nucleus for a centre of mass frame energy given by:

$$E_{\text{c.m.}}(\text{exit}) = E_{\text{c.m.}}(\text{entrance}) + Q$$

where $Q = Q_{\text{gg}} - E_{\text{ex}}$ (N.B. Q is often negative)

Thus one should use a separate set of parameters for each exit channel as E_{ex} varies. However, this is often ignored and a single set used. If E_{ex} is large (it can be several MeV) this may not be a good approximation

Also, while one can always measure the entrance channel elastic scattering this is sometimes not feasible (or even possible if one of the reaction partners is unbound) for the exit channel

We are thus often constrained to use potential parameters that describe the elastic scattering of similar systems at similar energies to those we require or, particularly with light-ion reactions, to use global parameter sets.

It is best to avoid this where possible, so when planning experiments it is good practice to allow for a measurement of the elastic scattering in the entrance channel at least

The other ingredients we need are the so-called overlap functions between the projectile-like and target-like nuclei: for the reaction $A(a,b)B$ where, for example, $a = x + b$, $B = x + A$, these are the wave functions represented by:

$$\langle a | x + b \rangle \text{ and } \langle B | x + A \rangle$$

To make this less abstract, we will consider the reaction $^{12}\text{C}(d,p)^{13}\text{C}$. In this case we need the overlap functions:

$$\langle d | n + p \rangle \text{ and } \langle ^{13}\text{C} | n + ^{12}\text{C} \rangle.$$

How do we obtain them?

FRESCO calculates them, but we must provide some information. For each overlap we need to define a potential to bind the transferred particle – here the neutron – onto the “core”, in this case the proton or the ^{12}C

These potentials are purely real (no absorption required) and the choice of form is to some extent arbitrary, although limits are placed by what is physically reasonable

They are often of Woods-Saxon form, and a popular – one might almost say “standard” – choice of parameters is:

$$R = 1.25 \times A^{1/3} \text{ fm}, a = 0.65 \text{ fm},$$

with the well depth (V) adjusted to yield the required binding energy; sometimes called the “well depth prescription”

These parameters usually give physically reasonable results for overlaps with a heavy ion as core. For light-ion overlaps like $\langle d | n + p \rangle$ there are better choices available such as the Reid soft-core potential which give realistic wave functions

Finally, to enable FRESKO to calculate the wave functions for the overlaps we need to define some quantum numbers.

Firstly, the intrinsic spin, s , of the transferred particle, the neutron in our example, so $s = 1/2$.

Secondly, some characteristics of the “transferred particle + core” wave function of the composite nuclei; in our example the $n + p$ wave function of the deuteron and the $n + {}^{12}\text{C}$ wave function of the ${}^{13}\text{C}$.

These are the three numbers N , L and J . N is the number of nodes in the wave function (we shall return to this shortly), L the angular momentum of the transferred particle relative to the core and J the total spin ($\mathbf{J} = \mathbf{L} + \mathbf{S}$) of the transferred particle

If the core has spin zero, determining L and J is particularly simple. We will stick to our example and see how they are defined for the $n + {}^{12}\text{C}$ overlap of the $1/2^-$ ground state of ${}^{13}\text{C}$.

We obtain L and J via (LS) J coupling. We want to end up with a state of spin $1/2$. Since the ${}^{12}\text{C}$ core has spin zero and the neutron spin $1/2$, one would naturally think of $L=0$...

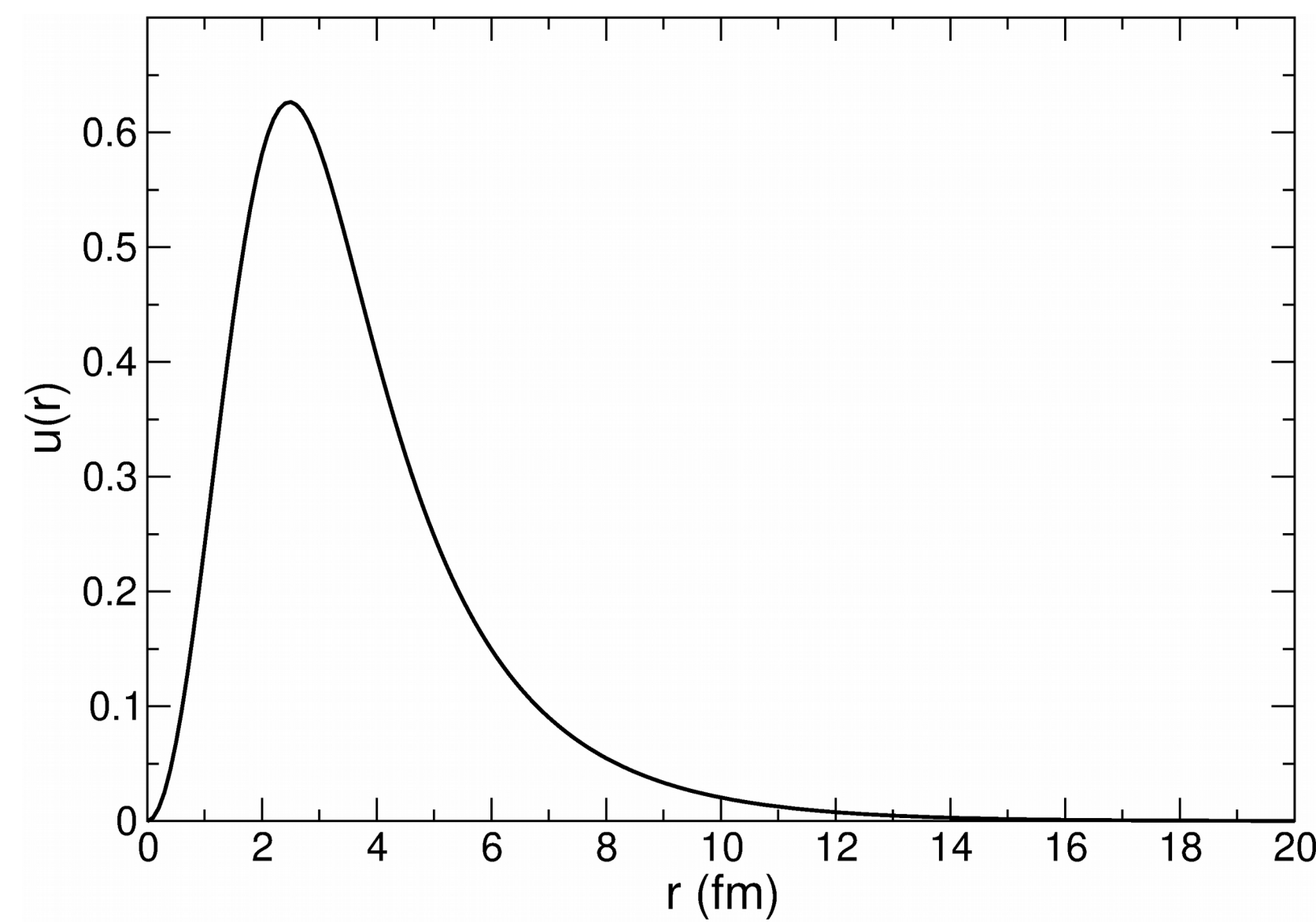
However, the parity of the ground state of ^{13}C is negative, so L must be odd: $L=0$ will not do. Quantum mechanically the vector sum $\mathbf{J} = \mathbf{L} + \mathbf{S}$ with $\mathbf{J} = \mathbf{1/2}$ and $\mathbf{S} = \mathbf{1/2}$ means that L can therefore only be 1

In terms of the shell model, our transferred neutron fills a vacant $1p_{1/2}$ orbital outside the inert (in our model) ^{12}C core. This brings us neatly to how we determine N

N is the number of nodes in the $n + ^{12}\text{C}$ radial wave function. For nucleon transfer, this is also the principal quantum number of the orbital the transferred particle is placed in with respect to the (inert) core

We therefore see that $N = 1$. A word of caution; there are two standard definitions of N in the literature so we need to make sure we use the right one.

All bound state wave functions contain at least two nodes, one at $r = 0$ and one at $r = \infty$:



Neither convention counts the node at $r = \infty$, but one does count the node at $r = 0$ (that used by FRESKO) while the other does not (that used by, e.g., DWUCK)

Under the DWUCK convention we would have $N=0$ for the ground state of ^{13}C , whereas in FRESKO it is $N=1$. Thus, in FRESKO $N \geq 1$. The value of N will affect the magnitude of the transfer cross section, so it is important to use the correct convention!

Things are slightly more complicated when the core nucleus has non-zero spin, as for the $\langle d | n + p \rangle$ overlap, but we will leave that for the exercises tomorrow.

Finally, we also need to supply a spectroscopic factor (in fact, two spectroscopic factors) to normalise our calculation to data. We will return to this later.

You are probably rather impatient by now to see how all this is done in practice, so without further ado let us look at a typical FRESCO DWBA input for the $^{12}\text{C}(\text{d},\text{p})^{13}\text{C}$ reaction to the ground state of ^{13}C

```

12C(d,p)13C at 30 MeV DWBA
0.10      30.0      0.20      0.20      10.0      -0.8
  00. 20.  +.00      F F
2  0.0      180.      0.5      1
0.0      1      1 1 0      48
  1 1 0 0 1 3 0 0 0 1 0 0 1
2H      2.0141      1.0      1 12C      12.0000      6.0      0.0000
1.0      +1 0.0      1 0.0      +1 0.00
1H      1.0078      1.0      1 13C      13.0034      6.0      2.7217
0.5      +1 0.0      2 0.5      -1 0.0

      1 0 0      12.0      0.0      1.3
      1 1 0      110.7      0.9      0.801
      1 2 0      9.2      1.42      0.794
      1 3 0      4.66      0.9      0.551
      2 0 0      13.0      0.0      1.122
      2 1 0      54.57      1.122      0.57
      2 2 0      8.66      1.122      0.5
      2 3 0      5.5      1.122      0.57
      3 0 0      1.00      1.25
      3 1 5      1.00      1.00
      3 3 5      1.00      1.00
      3 4 5      1.00      1.00
      3 7 5      1.00      1.00
      4 0 0      12.0      0.0      1.25
      4 1 0      50.0      1.25      0.65
      4 3 0      6.0      1.25      0.65
0
      1 2 1 2-1 3      1 0 2 0.5 1 1.5 1 3 0 2.2246 0 3 0
      3      1 2-2 0      1 1      0.5      0.5      4 0 4.9463 1 1 0

      -2      1      7 0-1 0
      1      1      1      1      1.0
      -2      1      1      3      0.8775
      0      1      1
30.0
EOF

```

Card 13

Card 15

Card 17

Most of this will look familiar, but there are three types of cards – 13, 15 and 17 – you have not seen before, plus some new variables on card 1.

We will look at the cards 13 first. The 1st card of this type defines the $\langle d | n + p \rangle$ overlap calculated using the Reid soft core potential [R. V. Reid, jr., Ann. Phys. (N.Y.) **50**, 441 (1968)]:

1 2 1 2-1 3 1 0 2 0.5 1 1.5 1 3 0 2.2246 0 3 0

The details of this particular overlap are beyond the scope of this introduction, but we include it as an example of how a more sophisticated overlap function can be calculated.

The $\langle {}^{13}\text{C} \mid n + {}^{12}\text{C} \rangle$ case is more straightforward:

3 1 2-2 0 1 1 0.5 0.5 4 0 4.9463 1 1 0

Section 2.5 of the FRESKO manual gives the details but we will also go through them now:

KN1=3 is merely a label and KN2 is not used here

IC1, IC2 and IN define the overlap

KIND=0 denotes use (LS)J coupling (standard for transfers)

NN, LN, SN and JN are N, L, S and J of the transferred particle

KBPOT is the label of the binding potential (number 4 here)

BE is the binding energy of the transferred particle to the core (in MeV)

We now come to card 15 and cards 17 (the minimum number of the latter is 2 for a finite-range transfer). These define the actual couplings:

-2	1	7	0	-1	0
	1	1	1	1	1.0
	-2	1	1	3	0.8775

Section 2.6 of the manual deals with these cards.

On card 15, the initial negative sign means one-way coupling only (i.e. DWBA). From partition 1 to partition 2. The 7 (variable KIND) denotes finite-range transfer

The next three numbers, IP1, IP2 and IP3 define the interaction.

IP1 = 0 indicates POST form DWBA (most accurate for stripping calculations; PRIOR is best for pickup)

IP2 = -1 denotes calculate the full complex remnant term

IP3 defines the core-core interaction (in this case the $p + {}^{12}\text{C}$ interaction). IP3=0 means use the $p + {}^{13}\text{C}$ optical potential (standard convention).

Now cards 17, which define the couplings. $IN1 = 1$ denotes projectile overlap, $IN1 = 2$ target overlap.

IB denotes the state of the composite nucleus (d or ^{13}C here)

IA denotes the state of the core nucleus (p or ^{12}C here)

KN is the label of the form factor

A is the spectroscopic *amplitude*, i.e. the square root of the spectroscopic factor C^2S . In DWBA the sign of A is not important.

A word on the binding energy (B.E.). In our example, the value is easily obtained. The core (^{12}C) and composite (^{13}C) nuclei are both in their ground states, so the B.E. of the transferred neutron is simply equal to the single neutron removal energy of ^{13}C , S_n

For stripping reactions leading to excited states of the composite nucleus (^{13}C) of excitation energy E_{ex} and with the ^{12}C core still in its ground state: $\text{B.E.} = S_n - E_{\text{ex}}$.

For pickup from the composite in its ground state to the core in an excited state: $\text{B.E.} = S_n + E_{\text{ex}}$

This is fine for bound states, but what to do if B.E. becomes negative?

In this case we are dealing with transfer to a resonance or a virtual state, which as such will have a finite width. If the level is close to the relevant emission threshold and is reasonably narrow, then we may use the weak binding energy approximation

In this approximation we use the correct excitation energy to define the kinematics (i.e. on card 7) but to calculate the form factor we use a small binding energy, say $B.E. = 0.01 \text{ MeV}$

It is possible to take the finite width into account and use the actual (negative) B.E. but this does not usually give a stable numerical result in finite-range DWBA

In analyses of transfer data the goal is often to extract C^2S . The magnitude of the DWBA cross section is linearly proportional to the value of C^2S , which amounts to a normalisation factor between theory and experiment

In finite-range DWBA the magnitude of the cross section actually depends on the *product of the projectile and target overlap spectroscopic factors*, and this product is determined by normalising the theoretical cross section to the data.

However, particularly in light-ion reactions, the projectile overlap is often more or less well known, from structure theory or previous experiments, so we can extract a C^2S value for the target overlap

Points to be aware of:

FRESCO uses the spectroscopic *amplitude*, the square root of C^2S .

The definition of the spectroscopic factor in FRESCO includes the *isospin* Clebsch-Gordon coefficient C – not always the case

In our example, for the $\langle ^{13}\text{C} \mid n + ^{12}\text{C} \rangle$ overlap C is the following isospin Clebsch-Gordon coefficient:

$$\langle T(^{12}\text{C}), T_z(^{12}\text{C}); T(n), T_z(n) \mid T(^{13}\text{C}), T_z(^{13}\text{C}) \rangle = \\ \langle 0, 0; 1/2, 1/2 \mid 1/2, 1/2 \rangle = 1.0$$

Returning to our input file, let us now look at card 1:

0.10 30.0 0.20 0.20 10.0 -0.8

The first three variables are familiar: HCM, RMATCH and RINTP. HCM and RMATCH have the same functions as previously. RINTP is a step size for the kernel function and is always a multiple of HCM: $2 \times \text{HCM}$ is usually reasonable

The next three are specific to (finite-range) transfer reactions: HNL, RNL and CENTRE. Their functions are discussed in the introduction to the FRESCO manual

HNL is a step size, either a fraction or a multiple of HCM

RNL is the non-local range of the kernel function, centred at CENTRE and calculated at intervals of HNL. How are these values chosen?

For light ions HNL is recommended to be a multiple of HCM:
 $HNL = HCM$ or $HNL = 2 \times HCM$ are reasonable values. For
heavy ions, $HNL = 0.5 \times HCM$ is a good starting point.

RNL and CENTRE are set by trial and error: in fact,
FRESCO gives suggestions as to the required values in the
main output file ...

```

0RECOMMENDED NON-LOCAL WIDTH IS GREATER THAN      4.80 FM.,  RECOMMENDED CENTRATION  -0.80, after      0.07 secs.
0Relative non-local usages (rms) are
  0.000   0.000   0.000   0.000   0.000   0.001   0.001   0.002   0.003   0.006   0.011   0.018   0.029   0.046   0.069
  0.098   0.135   0.176   0.221   0.267   0.313   0.366   0.459   0.719   1.575   4.394  13.238  36.731  62.330 100.000
55.423  42.119  16.924   5.023   1.347   0.357   0.105   0.047   0.030   0.022   0.016   0.011   0.008   0.006   0.004
  0.003   0.002   0.002   0.001   0.001   0.001

```

Provided the range is from ≈ 0.000 to ≈ 0.000 , RNL and
CENTRE are sufficient

As well as the cross section angular distributions, contained in the file fort.16 in the order they are given in the input file, FRESCO can also provide us with other useful quantities.

In the main output file, you will see a section like this:

```
OCUMULATIVE REACTION cross section          = 1026.27257  <L> =      5.62  <L**2> =      37.9
OCUMULATIVE outgoing cross sections in partition 1 :      0.00000
OCUMULATIVE outgoing cross sections in partition 2 :      8.65904
OCumulative ABSORBTION by Imaginary Potentials = 1017.61353  <L> =      5.63  <L**2> =      38.0
  Fusion for specific 2H      M-states : 1018.967471 1014.905643 1018.967471
OCUMULATIVE OUTGOING cross section          =      8.65904
```

“CUMULATIVE REACTION cross section” is the reaction cross section in mb, which you are familiar with from part 1.

“CUMULATIVE outgoing cross sections in partition 2” denotes the integrated cross section in mb for each state coupled to in partition 2, in the order in which they are given in the input file

“Cumulative ABSORPTION by Imaginary Potentials” is just that, the summed cross section in mb produced by the imaginary parts of the optical potentials in all channels.

Finally, “CUMULATIVE OUTGOING cross section” is the sum of all the integrated cross sections for all channels that are coupled to in all partitions, again in mb.

FRESCO can also calculate the ANC. In fact, FRESCO will calculate the so-called single-particle ANC which must then be multiplied by the corresponding spectroscopic amplitude. The result may then be squared to give the usual squared ANC, C^2 , often referred to simply as “the ANC” in the literature (the standard nomenclature is unfortunate ...)

Where are the single-particle ANCs for each of the overlaps displayed in the output?

In the output file of our example, for the $\langle ^{13}\text{C} \mid n + ^{12}\text{C} \rangle$ overlap you will see the following:

```

No.   P1 P2 IN KIND T N   L   S1 IA J/S   IB BIND XFER   BE       SC PC FIL AFRAC Adjust to Z Mass   K       Norm   rms       D0       D   ANC/Gsp
At R=   29.900, ANC in ch 3 is   1.8494   ,   eta,eta*ln(2*rho)+rho =   0.0000       14.031
3:   1   2   2   0       1   1 0.5   1 0.5   0   4   0   4.9463   1   1   0 0.0000 VR   0.8873; 0. 1.009 0.4693 1.0000   3.198 -128.7 -177.7   1.8494

```

where the single-particle ANC is $1.8494 \text{ fm}^{-1/2}$. The radius used to normalise the bound state radial wave function to the Whittaker function is also given, 29.9 fm here.

More detail is given in file fort.46, where you will find something like this:

```

&
# For Channel # 3: l = 1, j = 0.5, c# 1; k,eta = 0.46926 0.00000
  0.900   0.910958E-01   0.201106   2.20763   -1.25045   0   0.655514
  1.000   0.122827     0.240537   1.95833   -1.14988   0   0.625464
  1.100   0.160311     0.281015   1.75294   -1.06885   0   0.596791
.
.
.
 29.700   1.84935     0.175516E-05   0.949067E-06   -0.471516   0   0.885529E-06
 29.800   1.84935     0.167432E-05   0.905356E-06   -0.471501   0   0.844934E-06
 29.900   1.84935     0.159721E-05   0.863660E-06   -0.471487   0   0.806201E-06
&

```

The first line gives details of the state in question: the L and J values of the transferred particle with respect to the core and the wave number or momentum (k) and Sommerfeld parameters (η) for the $n + {}^{12}\text{C}$ relative motion (here η is zero since the neutron is uncharged)

The columns represent the radius, the “ANC” (i.e. the ratio of the radial wave function to the Whittaker function), the radial wave function $u(r)$ and the Whittaker function itself (the other columns do not concern us).

This can be useful if you want to plot $u(r)$ and is also a useful check on how quickly $u(r)$ matches onto the Whittaker function, i.e. the minimum radius to define a stable ANC

We have thus far dealt with nucleon (specifically, neutron, but the principle is the same for proton) transfer. We shall now give an example of cluster transfer by considering the $^{58}\text{Ni}(^7\text{Li},t)^{62}\text{Zn}$ α -cluster transfer reaction

We are now dealing with a transfer reaction induced by a heavy ion projectile but, as with the optical model, the input is essentially the same apart from a few details concerning “control” parameters

In the following input file I have included transfer to two states in ^{62}Zn , mainly as an example of how this is done. In DWBA it is immaterial whether we couple to more than one state simultaneously in the same input file; the result is the same if we produce two individual inputs

```

7Li + 58Ni at 34 MeV
0.06      25.0      0.12      0.03      8.0      -0.18
  00. 60.  +.00      F F
3      0.0      180.      0.5      1
0.0      1      1 1 0      48
  1 1 0 0 1 3 0 0 0 1 0 0 1
7Li      7.0160      3.0      1      58Ni      57.9353 28.0      0.0
1.5      -1 0.0000      1      0.0      +1 0.00
3H      3.0160      1.0      2      62Zn      61.9343 30.0      0.9022
0.5      +1 0.0000      2      0.0      +1 0.00
          1      5      2.0      +1 0.94

```

```

  1 0 0      58.0      0.0      1.3
  1 1 0      114.2      1.286      0.853      23.48      1.739      0.809
  2 0 0      62.0      0.0      1.3
  2 1 0      159.2      1.20      0.72      31.64      1.40      0.84
  2 3 0      2.5      1.20      0.72
  3 0 0      4.0      0.0      1.9497
  3 1 2 83.5706      0.0      2.52
  4 0 0      58.0      0.0      1.23
  4 1 0      50.0      1.23      0.65
  5 0 0      62.0      0.0      1.3
  5 1 0      159.4      1.20      0.72      31.97      1.40      0.84
  5 3 0      2.5      1.20      0.72
0
  1      1 2 1 0      2 1      0.0      1.0      3 0      2.4670      1 0 0
  2      1 2 2 0      7 0      0.0      0.0      4 0      3.3692      1 0 0
  3      1 2 2 0      6 2      0.0      2.0      4 0      2.4292      1 0 0

-2      1      7 0-1 0
          1      1      1      1      1.0000
          2      1      1      2      0.3241
          -2      2      1      3      0.1702
0      1      1
34.0
EOF

```

This should look familiar since it is not significantly different from the previous $^{12}\text{C}(\text{d},\text{p})^{13}\text{C}$ example. Note that HNL is now a fraction of HCM, as recommended for heavy-ion reactions. However, $\text{HNL} = \text{HCM}$ is usually sufficient.

Other points to note are that the $\alpha + \text{t}$ binding potential ($\text{KP} = 3$) is of Gaussian form ($\text{SHAPE} = 2$) rather than Woods-Saxon. The corresponding spectroscopic amplitude also happens to be 1.0 – this is not always the case for projectile overlaps!

Choice of binding potential radius for cluster overlaps is often problematic; there is no agreed formula and it is best practice to fix it using additional information, either theoretical or experimental, if possible

Finally, you may have noticed that the values of N for the two target-like overlaps are large, 7 and 6 for the ^{62}Zn 0^+ and 2^+ state respectively. These values cannot, of course, be simple shell model principal quantum numbers, so how were they obtained?

For cluster transfer, deriving the value of N is somewhat more complicated. One way of doing so – the most commonly used – is the so-called Talmi-Moshinsky relation:

$$2(N - 1) + L = \sum 2(n_i - 1) + \ell_i$$

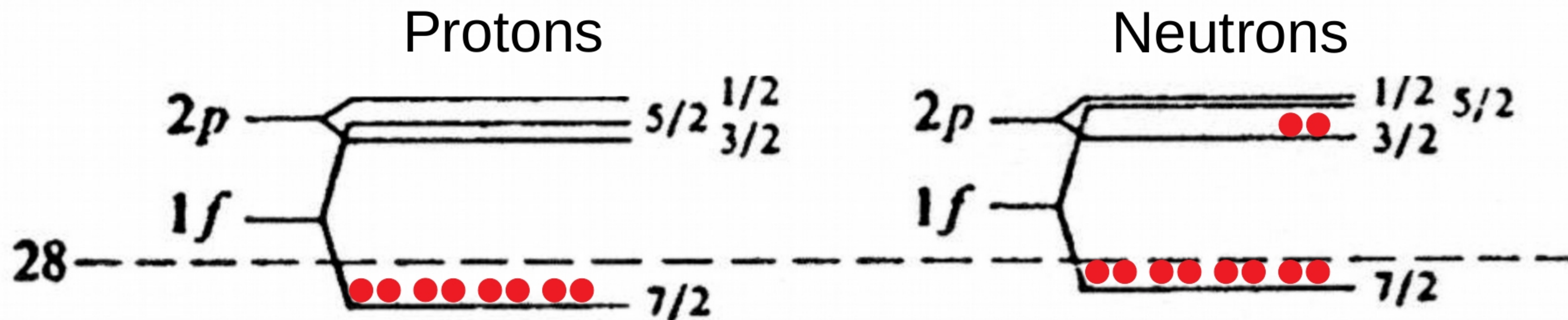
there are some extra terms for transfer of particles larger than the α particle, but it is unlikely you will need them

The N and L are the number of nodes in the radial wave function for the cluster-core system and the angular momentum of the cluster relative to the core.

The n_i and ℓ_i are the principal quantum number and angular momentum of the shell model levels in which the individual nucleons that go to make up the cluster are placed, the sum i running over the number of nucleons (here $i = 4$).

How do we fix N ? First, we decide on the value of L . For transfer of a 0^+ cluster (the α particle) onto a 0^+ core (^{58}Ni) to produce a composite nucleus (^{62}Zn) in a 0^+ state the quantum mechanical rules of angular momentum addition tell us that L can only be equal to zero

The ^{58}Ni core has 28 protons and 30 neutrons, so the available shell model levels appear as follows (we denote the final few protons and neutrons in ^{58}Ni by filled red circles)



Thus, adding an α particle the additional 2 protons and 2 neutrons would naturally go into the $2p_{3/2}$ orbitals to form ^{62}Zn in its 0^+ ground state. We therefore have that:

$$2(N - 1) + 0 = 2(2 - 1) + 1 + 2(2 - 1) + 1 + 2(2 - 1) + 1 + 2(2 - 1) + 1$$

so that $2N - 2 = 12$ and $N = 7$. Similarly, for the ^{62}Zn 2^+ state we get $N = 6$

Note that for cluster transfers the same final level may have more than one component with different N values but the same L and J . These components may correspond, for example, to different shell model configurations that can contribute to the final cluster state

However, inclusion of this level of detail really requires a theoretical calculation for the state(s) of interest using some sort of structure model. Trying to extract this sort of information empirically will usually simply result in increasing the number of variables to be determined from a limited data set

Finally, we come to the use of coupled reaction channels (CRC) for the calculation of transfer reactions. This is really very simple in FRESCO, since it requires only a few minor modifications of the DWBA input.

However, just as with CC for inelastic scattering, CRC includes so-called “back coupling” effects which can alter the elastic scattering. If the transfer couplings are strong you may need to re-tune the entrance channel optical potential parameters to recover the fit to the elastic scattering

What does a CRC input look like? Returning to our $^{12}\text{C}(\text{d},\text{p})^{13}\text{C}$ example ...

```

→ 12C(d,p)13C at 30 MeV CRC
0.10 30.0 0.20 0.20 15.0 -3.5
00. 20. +.00 F F
2 0.0 180. 0.5 1
→ 0.001 1 50 1 0 48
1 1 0 0 1 3 0 0 0 1 0 0 1
2H 2.0141 1.0 1 12C 12.0000 6.0 0.0000
1.0 +1 0.0 1 0.0 +1 0.00
1H 1.0078 1.0 1 13C 13.0034 6.0 2.7217
0.5 +1 0.0 2 0.5 -1 0.0

1 0 0 12.0 0.0 1.3
1 1 0 110.7 0.9 0.801
1 2 0 9.2 1.42 0.794
1 3 0 4.66 0.9 0.551
2 0 0 13.0 0.0 1.122
2 1 0 54.57 1.122 0.57
2 2 0 8.66 1.122 0.5
2 3 0 5.5 1.122 0.57
3 0 0 1.00 1.25
3 1 5 1.00 1.00
3 3 5 1.00 1.00
3 4 5 1.00 1.00
3 7 5 1.00 1.00
4 0 0 12.0 0.0 1.25
4 1 0 50.0 1.25 0.65
4 3 0 6.0 1.25 0.65
0
1 2 1 2-1 3 1 0 2 0.5 1 1.5 1 3 0 2.2246 0 3 0
3 1 2-2 0 1 1 0.5 0.5 4 0 4.9463 1 1 0
→ 2 1 7 0-2 0
1 1 1 1 1.0
-2 1 1 3 0.8775
0 1 1
30.0
EOF

```

Cards where there have been changes compared to the input for DWBA are marked with arrows: that really is all there is to it!

All well and good, but why have these changes been made and what do they do?

Taking card 4 first we now have:

0.001 1 50 1 0 48

ITER is now 50, not 1 and IPS 0.001, not 0.0. For transfer channels, FRESCO solves CRC by iteratively performing DWBA calculations.

IT0 = 0 and ITER = 50 mean do at least 1 iteration up to a maximum of 50, with IPS = 0.001 meaning “stop when the difference between the S matrix elements of successive iterations is less than 0.001%”

Exact values are not that important provided you allow enough iterations for convergence

Now card 15:

2 1 7 0-2 0

Here there are two changes:

ICTO is no longer negative. This means allow two-way couplings (essential for CRC)

IP is now -2. This means that, in addition to calculating the full complex remnant term also calculate the non-orthogonality correction (again, essential for CRC)

Finally, on Card 1 we have increased RNL to 15.0 fm and altered CENTRE to -3.5 fm. Why is this?

If we look at the output file we find a new section connected with the non-orthogonality correction:

```
So NON-ORTHOGONALITY SUPPLEMENT for an IC7=0 (POST) coupling

A real*8 is          8  bytes
NLL,NLN,NLO =        150          151          76
0File 11 needs RL =   11400    REAL*8 numbers, and NR =    55. i.e.    5.02 Megabytes
File          11 RECL =      91200
0RECOMMENDED NON-LOCAL WIDTH IS GREATER THAN 15.00 FM., RECOMMENDED CENTRATION -3.50, after 0.05 secs.
0Relative non-local usages (rms) are
  0.043  0.049  0.055  0.062  0.070  0.079  0.089  0.101  0.114  0.129  0.146  0.166  0.188  0.214  0.243
  0.276  0.314  0.358  0.408  0.465  0.532  0.608  0.697  0.799  0.918  1.055  1.216  1.402  1.621  1.877
  2.178  2.532  2.951  3.447  4.036  4.739  5.579  6.585  7.794  9.250 11.004 13.121 15.674 18.751 22.457
 26.909 32.246 38.628 46.233 55.252 65.830 77.858 90.282 99.612 100.000 91.512 82.969 72.181 56.125 39.426
 26.005 16.560 10.354  6.431  4.000  2.505  1.585  1.014  0.656  0.429  0.284  0.189  0.127  0.086  0.059
 0.040
```

We adjust RNL and CENTRE to ensure we cover the required range for the integration; FRESCO suggests values

In addition to coupling effects on the elastic scattering CRC differs from DWBA in two further aspects:

One should, at least in principle, couple all channels of importance simultaneously, i.e. in the same input file, since coupling effects are cumulative and are usually different from the sum of their parts

In CRC the (relative) signs of the spectroscopic amplitudes matter, due to interference effects (this is why FRESCO uses the spectroscopic amplitudes rather than the factors, which are positive definite)

The magnitude of the cross section for transfer to a given state is still, to a good approximation, linearly dependent on the spectroscopic factor in many (if not most) CRC calculations

This brings us to the end of the lectures concerning the practical application of direct reaction theory to the analysis of data. We have not covered the simultaneous inclusion of inelastic scattering and transfer reactions – using either CCBA or CRC + CC – nor have we addressed CDCC calculations.

However, one has to stop somewhere!

Thank you for your attention



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